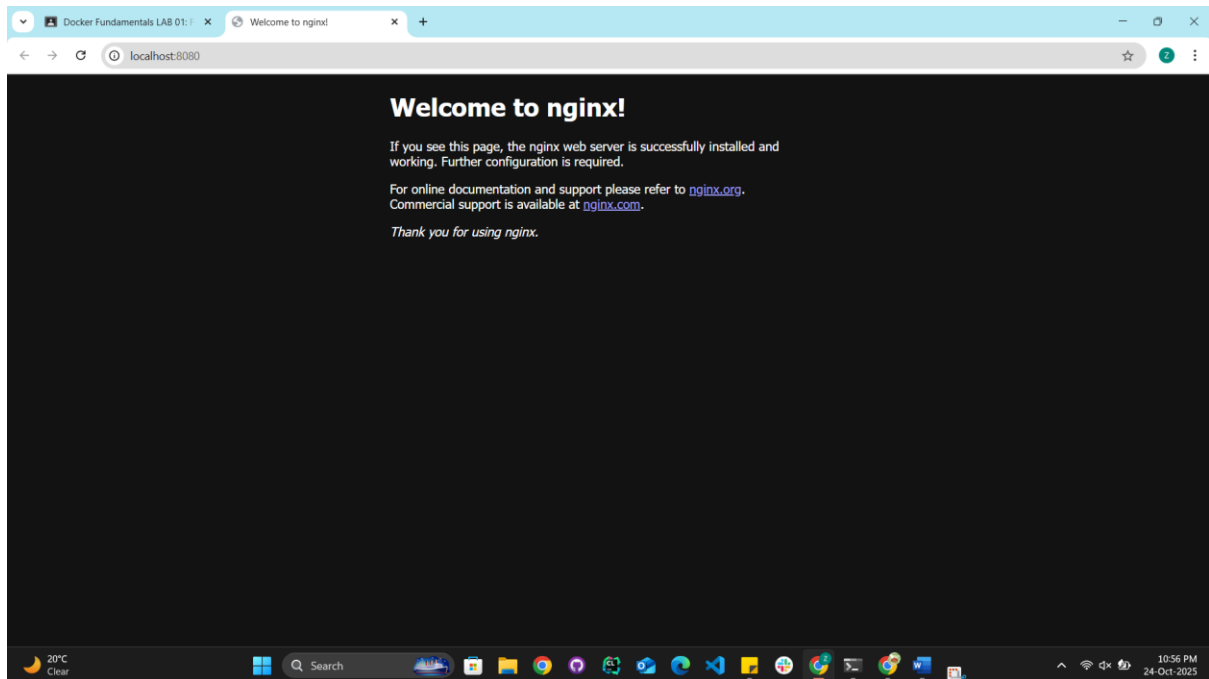


Screenshot 1:



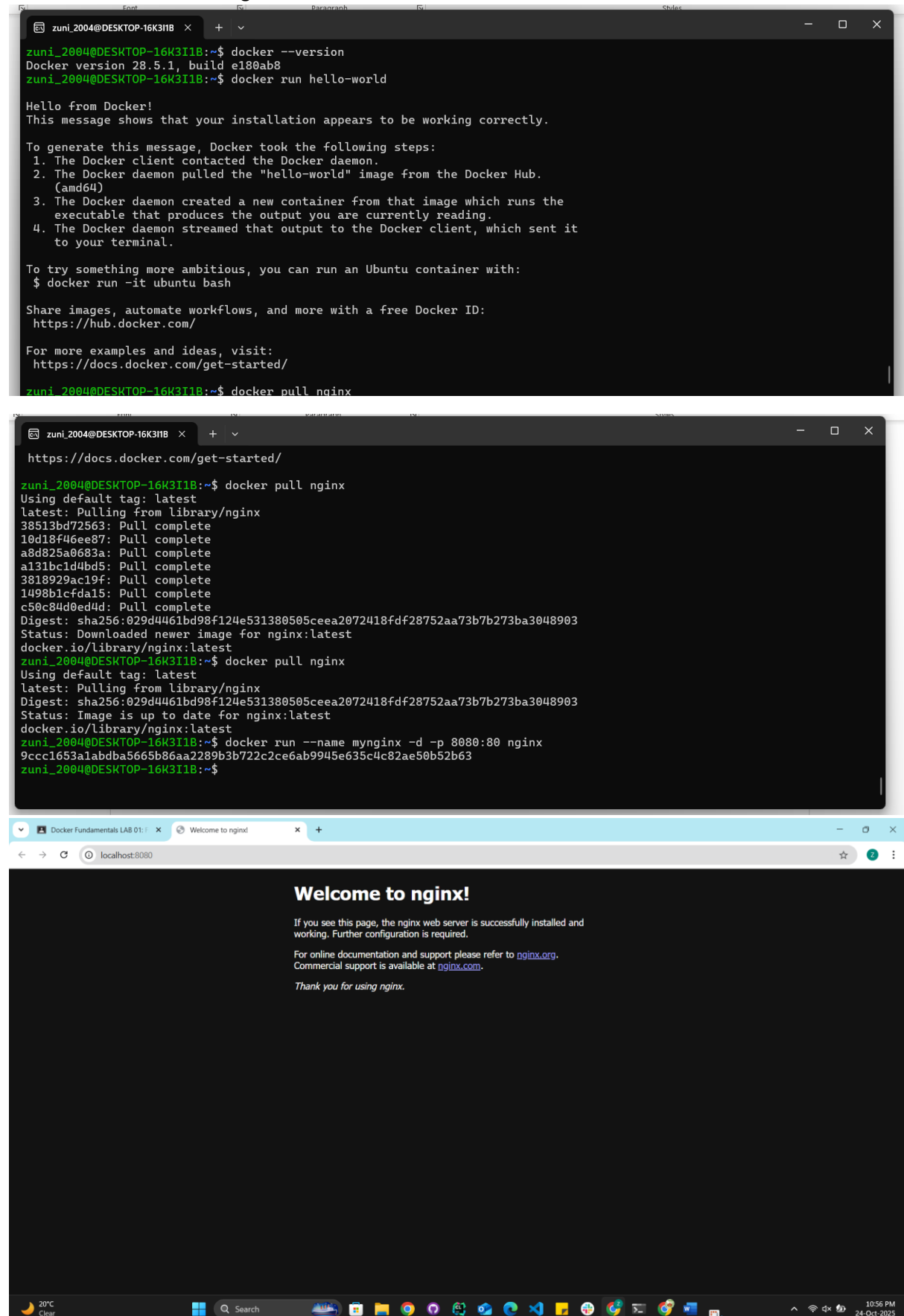
Screenshot 2:

```
zuni_2004@DESKTOP-16K311B:~$ mkdir mypythonapp
cd mypythonapp
zuni_2004@DESKTOP-16K311B:~/mypythonapp$ echo 'print("Hello from my first Docker container!")' > app.py
zuni_2004@DESKTOP-16K311B:~/mypythonapp$ nano Dockerfile
zuni_2004@DESKTOP-16K311B:~/mypythonapp$ docker build -t mypythonapp:v1 .
[+] Building 17.3s (8/8) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile                0.1s
=> => transferring dockerfile: 121B                                0.0s
=> [internal] load metadata for docker.io/library/python:3.9-slim 4.6s
=> [internal] load .dockerignore                                  0.1s
=> => transferring context: 2B                                       0.0s
=> [internal] load build context                                  0.1s
=> => transferring context: 80B                                       0.0s
=> [1/3] FROM docker.io/library/python:3.9-slim@sha256:545badebace9a958b98d3e272f0f0d46c0a1a389ac77e24c33f2e7b5 11.9s
=> resolve docker.io/library/python:3.9-slim@sha256:545badebace9a958b98d3e272f0f0d46c0a1a389ac77e24c33f2e7b54 0.0s
=> sha256:dc9f486a31b5c0642ed5606b41c820fae66e6d91003c32e4f3d1624b6e7425cd 1.74kB / 1.74kB 0.0s
=> sha256:725dd64d2f62808e0aee4d5d916f4c208275716e74cc700ed68aa17078411ca9 5.30kB / 5.30kB 0.0s
=> sha256:c57466e36deb99f355eab1131bc21d799709189ecdd7fb41cf4c17d5f53724e2 1.29MB / 1.29MB 10.2s
=> sha256:6f010d693edcb2d965ff3c2a741fc92cded69746aa616f746e7813896fdd443b 13.87MB / 13.87MB 7.5s
=> sha256:21ed5e7e522581efdab14b4fde1d7d493901b4d25736b947520695c7911da66b 250B / 250B 1.6s
=> sha256:545badebace9a958b98d3e272f0f0d46c0a1a389ac77e24c33f2e7b548ce1b6b 10.36kB / 10.36kB 0.0s
=> extracting sha256:c57466e36deb99f355eab1131bc21d799709189ecdd7fb41cf4c17d5f53724e2 0.2s
=> extracting sha256:6f010d693edcb2d965ff3c2a741fc92cded69746aa616f746e7813896fdd443b 1.0s
=> extracting sha256:21ed5e7e522581efdab14b4fde1d7d493901b4d25736b947520695c7911da66b 0.0s
=> [2/3] COPY app.py /app/app.py                                  0.1s
=> [3/3] WORKDIR /app                                             0.1s
=> exporting to image                                             0.1s
=> => exporting layers                                             0.1s
=> writing image sha256:b23b540ee5701746e391a044b1ba84efd3133f7fa9f9d103a29d5ae0a7891fc5 0.0s
=> naming to docker.io/library/mypythonapp:v1                    0.0s
zuni_2004@DESKTOP-16K311B:~/mypythonapp$ docker run --name pycontainer mypythonapp:v1
Hello from my first Docker container!
zuni_2004@DESKTOP-16K311B:~/mypythonapp$
```

Screenshot 3:

```
zuni_2004@DESKTOP-16K311B:~/mypythonapp$ docker ps -a
CONTAINER ID   IMAGE          COMMAND                  CREATED              STATUS              PORTS              NAMES
06700b98829d   hello-world    "/hello"                 27 minutes ago      Exited (0) 27 minutes ago      -                  elegant_blackwell
1f4fcd21344a   hello-world    "/hello"                 28 minutes ago      Exited (0) 28 minutes ago      -                  epic_jones
92d551d801e2   hello-world    "/hello"                 29 minutes ago      Exited (0) 29 minutes ago      -                  serene_bouman
zuni_2004@DESKTOP-16K311B:~/mypythonapp$
```

Task 1: Run and validate nginx container



```
zuni_2004@DESKTOP-16K3I1B: ~$ docker --version
Docker version 28.5.1, build e180ab8
zuni_2004@DESKTOP-16K3I1B: ~$ docker run hello-world

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
1. The Docker client contacted the Docker daemon.
2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
   (amd64)
3. The Docker daemon created a new container from that image which runs the
   executable that produces the output you are currently reading.
4. The Docker daemon streamed that output to the Docker client, which sent it
   to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
$ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
https://hub.docker.com/

For more examples and ideas, visit:
https://docs.docker.com/get-started/
zuni_2004@DESKTOP-16K3I1B: ~$ docker pull nginx

https://docs.docker.com/get-started/

zuni_2004@DESKTOP-16K3I1B: ~$ docker pull nginx
Using default tag: latest
latest: Pulling from library/nginx
38513bd72563: Pull complete
10d18f46ee87: Pull complete
a8d825a0683a: Pull complete
a131bc1d4bd5: Pull complete
3818929ac19f: Pull complete
1498b1cfda15: Pull complete
c50c84d0ed4d: Pull complete
Digest: sha256:029d4461bd98f124e531380505ceea2072418fdf28752aa73b7b273ba3048903
Status: Downloaded newer image for nginx:latest
docker.io/library/nginx:latest
zuni_2004@DESKTOP-16K3I1B: ~$ docker pull nginx
Using default tag: latest
latest: Pulling from library/nginx
Digest: sha256:029d4461bd98f124e531380505ceea2072418fdf28752aa73b7b273ba3048903
Status: Image is up to date for nginx:latest
docker.io/library/nginx:latest
zuni_2004@DESKTOP-16K3I1B: ~$ docker run --name mynginx -d -p 8080:80 nginx
9ccc1653a1abdba5665b86aa2289b3b722c2ce6ab9945e635c4c82ae50b52b63
zuni_2004@DESKTOP-16K3I1B: ~$
```

Docker Fundamentals LAB 01: Welcome to nginx!

localhost:8080

Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.

20°C Clear 10:55 PM 24 Oct 2025

Task 2: Create a Python “Hello” app container

```
zuni_2004@DESKTOP-16K3I1B:~$ docker ps
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS        PORTS
9ccc1653alab   nginx     "/docker-entrypoint. ..." About a minute Up About a minute 0.0.0.0:8080->80/tcp, [::]:80
80->80/tcp      mynginx

zuni_2004@DESKTOP-16K3I1B:~$ docker logs mynginx
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2025/10/24 17:54:26 [notice] 1#1: using the "epoll" event method
2025/10/24 17:54:26 [notice] 1#1: nginx/1.29.2
2025/10/24 17:54:26 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2025/10/24 17:54:26 [notice] 1#1: OS: Linux 5.15.167.4-microsoft-standard-WSL2
2025/10/24 17:54:26 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2025/10/24 17:54:26 [notice] 1#1: start worker processes
2025/10/24 17:54:26 [notice] 1#1: start worker process 29
2025/10/24 17:54:26 [notice] 1#1: start worker process 30
2025/10/24 17:54:26 [notice] 1#1: start worker process 31
2025/10/24 17:54:26 [notice] 1#1: start worker process 32
2025/10/24 17:54:26 [notice] 1#1: start worker process 33
```

```
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2025/10/24 17:54:26 [notice] 1#1: using the "epoll" event method
2025/10/24 17:54:26 [notice] 1#1: nginx/1.29.2
2025/10/24 17:54:26 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2025/10/24 17:54:26 [notice] 1#1: OS: Linux 5.15.167.4-microsoft-standard-WSL2
2025/10/24 17:54:26 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2025/10/24 17:54:26 [notice] 1#1: start worker processes
2025/10/24 17:54:26 [notice] 1#1: start worker process 29
2025/10/24 17:54:26 [notice] 1#1: start worker process 30
2025/10/24 17:54:26 [notice] 1#1: start worker process 31
2025/10/24 17:54:26 [notice] 1#1: start worker process 32
2025/10/24 17:54:26 [notice] 1#1: start worker process 33
2025/10/24 17:54:26 [notice] 1#1: start worker process 34
2025/10/24 17:54:26 [notice] 1#1: start worker process 35
2025/10/24 17:54:26 [notice] 1#1: start worker process 36
172.17.0.1 -- [24/Oct/2025:17:54:32 +0000] "GET / HTTP/1.1" 200 615 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) Appl
eWebKit/537.36 (KHTML, like Gecko) Chrome/141.0.0.0 Safari/537.36" "-"
172.17.0.1 -- [24/Oct/2025:17:54:32 +0000] "GET /favicon.ico HTTP/1.1" 404 555 "http://localhost:8080/" "Mozilla/5.0 (W
indows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/141.0.0.0 Safari/537.36" "-"
2025/10/24 17:54:32 [error] 29#29: *1 open() "/usr/share/nginx/html/favicon.ico" failed (2: No such file or directory),
client: 172.17.0.1, server: localhost, request: "GET /favicon.ico HTTP/1.1", host: "localhost:8080", referer: "http://l
ocalhost:8080/"
zuni_2004@DESKTOP-16K3I1B:~$
```

```
2025/10/24 17:54:26 [notice] 1#1: using the "epoll" event method
2025/10/24 17:54:26 [notice] 1#1: nginx/1.29.2
2025/10/24 17:54:26 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2025/10/24 17:54:26 [notice] 1#1: OS: Linux 5.15.167.4-microsoft-standard-WSL2
2025/10/24 17:54:26 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2025/10/24 17:54:26 [notice] 1#1: start worker processes
2025/10/24 17:54:26 [notice] 1#1: start worker process 29
2025/10/24 17:54:26 [notice] 1#1: start worker process 30
2025/10/24 17:54:26 [notice] 1#1: start worker process 31
2025/10/24 17:54:26 [notice] 1#1: start worker process 32
2025/10/24 17:54:26 [notice] 1#1: start worker process 33
2025/10/24 17:54:26 [notice] 1#1: start worker process 34
2025/10/24 17:54:26 [notice] 1#1: start worker process 35
2025/10/24 17:54:26 [notice] 1#1: start worker process 36
172.17.0.1 -- [24/Oct/2025:17:54:32 +0000] "GET / HTTP/1.1" 200 615 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) Appl
eWebKit/537.36 (KHTML, like Gecko) Chrome/141.0.0.0 Safari/537.36" "-"
172.17.0.1 -- [24/Oct/2025:17:54:32 +0000] "GET /favicon.ico HTTP/1.1" 404 555 "http://localhost:8080/" "Mozilla/5.0 (W
indows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/141.0.0.0 Safari/537.36" "-"
2025/10/24 17:54:32 [error] 29#29: *1 open() "/usr/share/nginx/html/favicon.ico" failed (2: No such file or directory),
client: 172.17.0.1, server: localhost, request: "GET /favicon.ico HTTP/1.1", host: "localhost:8080", referer: "http://l
ocalhost:8080/"
zuni_2004@DESKTOP-16K3I1B:~$ docker stop mynginx
mynginx
zuni_2004@DESKTOP-16K3I1B:~$ docker rm mynginx
mynginx
zuni_2004@DESKTOP-16K3I1B:~$
```

Step 5: Build a custom Docker image

```
zuni_2004@DESKTOP-16K3I1B: ~$ mkdir mypythonapp
cd mypythonapp
zuni_2004@DESKTOP-16K3I1B:~/mypythonapp$ echo 'print("Hello from my first Docker container!")' > app.py
zuni_2004@DESKTOP-16K3I1B:~/mypythonapp$ nano Dockerfile
zuni_2004@DESKTOP-16K3I1B:~/mypythonapp$ docker build -t mypythonapp:v1 .
[+] Building 17.3s (8/8) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile              0.1s
=> => transferring dockerfile: 121B                             0.0s
=> [internal] load metadata for docker.io/library/python:3.9-slim 4.6s
=> [internal] load .dockerignore                                0.1s
=> => transferring context: 2B                                    0.0s
=> [internal] load build context                                0.1s
=> => transferring context: 80B                                   0.0s
=> [1/3] FROM docker.io/library/python:3.9-slim@sha256:545badebace9a958b98d3e272f0f0d46c0a1a389ac77e24c33f2e7b5 11.9s
=> => resolve docker.io/library/python:3.9-slim@sha256:545badebace9a958b98d3e272f0f0d46c0a1a389ac77e24c33f2e7b54 0.0s
=> => sha256:dc9f486a31b5c0642ed5606b41c820fae66e6d91003c32e4f3d1624b6e7425cd 1.74kB / 1.74kB 0.0s
=> => sha256:725dd64d2f62808e0aee4d5d916f4c208275716e74cc700ed68aa17078411ca9 5.30kB / 5.30kB 0.0s
=> => sha256:c57466e36deb99f355eab1131bc21d799709189ecdd7fb41cf4c17d5f53724e2 1.29MB / 1.29MB 10.2s
=> => sha256:6f010d693edcb2d965ff3c2a741fc92cded69746aa616f746e7813896fdd443b 13.87MB / 13.87MB 7.5s
=> => sha256:21ed5e7e522581efdab14b4fde1d7d493901b4d25736b947520695c7911da66b 250B / 250B 1.6s
=> => sha256:545badebace9a958b98d3e272f0f0d46c0a1a389ac77e24c33f2e7b548ce1b6b 10.36kB / 10.36kB 0.0s
=> => extracting sha256:c57466e36deb99f355eab1131bc21d799709189ecdd7fb41cf4c17d5f53724e2 0.2s
=> => extracting sha256:6f010d693edcb2d965ff3c2a741fc92cded69746aa616f746e7813896fdd443b 1.0s
=> => extracting sha256:21ed5e7e522581efdab14b4fde1d7d493901b4d25736b947520695c7911da66b 0.0s
=> [2/3] COPY app.py /app/app.py                                0.1s
=> [3/3] WORKDIR /app                                           0.1s
=> => exporting to image                                          0.1s
=> => exporting layers                                           0.1s
=> => writing image sha256:b23b540ee5701746e391a044b1ba84efd3133f7fa9f9d103a29d5ae0a7891fc5 0.0s
=> => naming to docker.io/library/mypythonapp:v1                0.0s
zuni_2004@DESKTOP-16K3I1B:~/mypythonapp$ docker run --name pycontainer mypythonapp:v1
Hello from my first Docker container!
zuni_2004@DESKTOP-16K3I1B:~/mypythonapp$
```

Task 3: Cleanup

```
zuni_2004@DESKTOP-16K3I1B:~/mypythonapp$ docker stop pycontainer
pycontainer
zuni_2004@DESKTOP-16K3I1B:~/mypythonapp$ docker rm pycontainer
pycontainer
zuni_2004@DESKTOP-16K3I1B:~/mypythonapp$ docker images
REPOSITORY      TAG         IMAGE ID      CREATED        SIZE
mypythonapp     v1          b23b540ee570 3 minutes ago  122MB
nginx           latest      657fdcd1c365 2 weeks ago   152MB
hello-world     latest      1b44b5a3e06a 2 months ago  10.1kB
zuni_2004@DESKTOP-16K3I1B:~/mypythonapp$
```